

Machong Paul

Java Software Engineer

✉ macpaul201@gmail.com ☎ 07064446227 📍 Lagos, Nigeria

🌐 <https://www.linkedin.com/in/machong-paul/> 🏠 github.com/sirpolux/

Profile

Results-driven **Java Backend Engineer** with 5+ years of software engineering experience specializing in secure, high-throughput **financial systems, trade engineering,** and **corporate lending infrastructure.** Expertise across **Java 17/21, Spring Boot microservices, high-concurrency transaction processing,** and **automated debt-recovery engine design.** Experienced in building secure **multi-party B2B approval architectures,** and integrating **artificial intelligence** with standard financial protocols. Expert in database **optimization, distributed caching,** and **cloud infrastructure.**

Skills

Programming & Backend — Java, Spring Boot, Spring MVC, Spring Security, RESTful APIs, Microservices, Object-Oriented Programming, SOLID Principles, Design Patterns, UML, DBML(Database Markup Language), PHP, Laravel, JavaScript, React.js

Financial Systems — Customer Onboarding, KYC, NIBSS, Core Banking, Payment Processing, Fund Transfers, Transaction Processing, Account Management, Mandates

Messaging & Integration — REST API Integration, Apache Kafka, RabbitMQ, Service-to-Service Integration, Third-Party API Integration

Security & Testing — JUnit 5, Mockito, OAuth2, Authentication, Authorization, Encryption, Token Management, Unit Testing, Integration Testing, Debugging

Tools & Methodologies — Git, GitHub, Agile, Scrum, Problem Solving, Team Collaboration

Database Management — MySQL, PostgreSQL, Microsoft SQL Server, MongoDB, Hibernate, JPA, Spring Data JPA, JdbcTemplate, Redis

Cloud & DevOps — Docker, Kubernetes, Microsoft Azure, Azure DevOps, AWS, Google Cloud, Red Hat OpenShift, CI/CD

AI — AI Automation (n8n), Vibe coding (Lovable, Claude)

Professional Experience

Software Engineer

2023 – 2026 | Lagos, Nigeria

Fundtron Capital Limited

- **Developed backend services supporting** a high-throughput multi-asset corporate trading and fixed-income settlement system handling Treasury Bills, Bonds, and Commercial Papers, driving a **40% reduction in manual execution trade errors.**
- **Engineered an asynchronous multi-party consent and role-delegation engine** for enterprise lending clients, ensuring strict transactional integrity across complex multi-shareholder corporate structures.
- **Integrated a natural language AI trading pipeline** utilizing custom LLM parsing APIs to capture unstructured conversational text from traders, programmatically extract trade parameters, and convert them into structured market deals with zero latency.
- **Constructed automated identity-onboarding workflows** incorporating NIBSS validation hooks for real-time BVN and NIN parsing, dropping client verification dropout rates by **18%.**
- **Integrated Redis for distributed caching and session management,** caching fast-moving system configurations, live currency conversion rates, and user portfolio states to prevent excessive database read spikes and pool saturation during high-traffic intervals.
- **Implemented a localized rule engine for tier-based financial operations,** automating risk evaluation procedures for multi-stage credit, risk management, and finance officer disbursement flows.

Java Backend Engineer

2021 – 2023 | Abuja, Nigeria

MTech Innovation HUB

- **Developed and deployed enterprise-grade REST APIs** using Spring Boot, Spring Data JPA, and MySQL, transitioning a legacy system into a scalable modular monolith framework.
- **Optimized analytical reporting throughput** via database pagination patterns, composite indexes, and non-blocking stream mapping to prevent application heap overhead during export executions.

- **Conducted high-concurrency baseline testing** simulating over **5,000 requests per second** using k6/JMeter, optimizing connection pools, and utilizing Testcontainers for isolated validation mapping.
- **Participated in code reviews** and followed Git-based development practices to maintain code quality and support effective team collaboration.

Java Developer

2020 – 2021 | Nasarawa, Nigeria

Itech Solutions

- Developed and integrated **RESTful API endpoints** to support core business functionality and enable seamless communication between frontend and backend systems.
- Wrote **unit and integration tests** for backend components to identify defects early and improve the reliability of application releases.
- Participated in **code reviews** and followed **Git-based** development practices to maintain code quality and support effective team collaboration.
- Applied **SOLID principles, object-oriented programming**, and established design patterns to produce clean, maintainable, and scalable code.

Projects

Fundtron — Corporate Asset Trading & Settlement Infrastructure Platform

- **Designed a unified automated settlement routing gateway** that programmatically batches and dispatches transaction instruction matrices to major settlement banks via encrypted, fail-safe SMTP channels.
- **Integrated an AI text-box translation pipeline** powered by custom generative intelligence APIs, allowing traders to input plain-language statements to trigger automatic deal mapping, validation, and real-time execution.
- **Implemented an enterprise-wide real-time Mark-to-Market (MTM) engine** providing real-time pricing feeds to monitor realized and unrealized P&L accounts across variable books.

Krevus — Business Financing & Shareholder B2B Lending Ecosystem

- **Engineered a comprehensive B2B loan orchestration system** integrating custom KYC capture frameworks, automated NIBSS identity pipelines (BVN/NIN), and virtual account allocation layers.
- **Built a multi-tier corporate role delegation framework** establishing strict boundary configurations separating financial Requester and Approver permissions across multiple shareholders.
- **Authored a multi-stage credit-officer approval pipeline** routing applications sequentially through Risk Management, Credit Analysis, and Treasury disbursement checkpoints with built-in audit trails.
- **Created a resilient background debit scheduler** using Spring Batch and customized database transaction schemas to handle automatic mandate sweeps for scheduled collections.

FinTrack — Financial Management & Analytics

- **Developed transaction processing and wallet management services** using Java and Spring Data JPA, mapping complex business rules for multicurrency income, expense tracking, and consistent account-to-account transfers.
- **Built cross-cutting AOP aspects to decouple core logic**, implementing custom annotations.
- **Optimized heavy transaction history endpoints** by enforcing server-side pagination, writing tailored DTO projections to reduce heap memory overhead, and implementing database composite indexes to eliminate table-scan performance drops.
- **Wrote comprehensive automated test suites and diagnosed load bottlenecks**, using Test containers for isolated MySQL repository verification, MockMvc for API request contract assertions, and tracking connection pool saturation using HikariCP properties under simulated multi-threaded execution.

Education

B.Sc Computer Science

2015 – 2019

National Open University of Nigeria

Certificates

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|---------------------------------|--------------------------------------|---|
| - Developer Certification | - Scrum Fundamentals Certified (SFC) | - Software Engineering and Leadership |
| - RESTful Services Fundamentals | - Relational Databases Certification | - Introduction to Business Intelligence |